PHOLIO

Platform Architecture & Product Specification

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Pholio is a premium dual-sided marketplace that transforms how creative talent connects with professional representation. This document provides a comprehensive technical and product specification for the platform, covering the value proposition, user flows, system architecture, AI infrastructure, design system, and business model.

1. Executive Summary

1.1 Problem Statement

The modeling and talent representation industry is characterized by a fundamental discovery gap operating at both ends of the market. Talented individuals — models, actors, and creative talent — lack the professional presentation infrastructure required to attract top-tier agencies. Simultaneously, agencies are forced to sift through high volumes of amateur, poorly organized submissions to find qualified talent. This bilateral inefficiency slows representation, raises acquisition costs, and depresses overall market quality.

1.2 Solution

Pholio is a premium dual-sided platform that addresses this inefficiency at the top of the funnel. By combining AI-powered onboarding, intelligent match scoring, and a structured application pipeline, Pholio standardizes talent presentation and streamlines agency discovery — without attempting to replace the operational back-office infrastructure agencies already use (e.g., Syngency, CDM).

Pholio's positioning is explicitly luxury: the aesthetic and experience are closer to editorial fashion than SaaS. The platform is designed to feel like a product that belongs in the same world as Vogue — not a LinkedIn for models.

1.3 Market Scope

Primary focus is on models, with the platform open to adjacent talent categories including actors, influencers, and other creative professionals. Geographic launch is US-first, with a long-term global expansion strategy.

1.4 Business Model

Pholio currently operates on a single revenue stream: the Studio+ subscription tier for talent. Agencies are onboarded at no cost and by invitation only, ensuring platform quality. The Studio+ tier unlocks a premium suite of tools detailed in Section 4.

2. Platform Architecture

2.1 System Overview

Pholio is built as a React Single Page Application (SPA) served via Netlify's static hosting, backed by a Node.js/Express monolith deployed as serverless Netlify Functions (AWS Lambda wrappers). The database layer uses Neon, a serverless PostgreSQL provider chosen for its auto-scaling behavior, branching support for dev/prod environments, and consumption-based pricing model.

Frontend Framework	React 19 (Vite build system)
Routing	React Router v7
State Management	TanStack Query (server state) + Context API (auth / global UI)
Styling	TailwindCSS + Framer Motion (cinematic animations)
Forms & Validation	React Hook Form + Zod schema validation
Backend Runtime	Node.js (Express monolith, /api/* routes)
Backend Deployment	Netlify Functions (serverless AWS Lambda wrappers)
Frontend Deployment	Netlify (static hosting + client-side routing)
Database	Neon (serverless PostgreSQL)
ORM / Query Builder	Knex.js (SQL migrations and interactions)
Authentication	Firebase Auth (Email/Password + Google OAuth)
Payments	Stripe (Studio+ subscriptions)
PDF Generation	Puppeteer

2.2 AI & ML Infrastructure

Pholio uses two distinct Groq-hosted language model endpoints for AI functionality, each scoped to a specific task:

Vision Model	Groq — Llama-3.2-11b-Vision-Preview
Vision Role	Analyzes uploaded talent photos during onboarding. Extracts physical attributes (hair color, eye color, build) and generates

	predictive estimates for body measurements (height, bust, waist, hips).
Inference Model	Groq — Llama-3-70b-8192
Inference Role	Calculates agency/board match scores and generates AI-driven recommendations for Studio+ users (bio refinement, photo selection, smart portfolio building).

2.3 Architecture Decisions

- Monolith-in-Serverless: The Express app handles all API routes under /api/* as a single deployable unit within Netlify Functions. This reduces operational complexity at the current scale while retaining the option to decompose into microservices as the platform grows.
- Neon PostgreSQL: Selected over traditional hosted Postgres for its serverless autoscaling, which aligns with Pholio's variable load profile at launch, and its branching feature that enables isolated dev/staging environments without duplicating infrastructure.
- Firebase Auth: Chosen to offload authentication complexity — session management, OAuth flows, and token verification — while maintaining full control over user data in Neon.
- Groq for Inference: Low-latency LLM inference is critical during onboarding (users are waiting in real time). Groq's hardware-accelerated inference provides the response speed required for a seamless, real-time conversational experience.

3. User Types & Access Model

3.1 Talent (Supply Side)

Talent users are the supply side of the marketplace — primarily models, with the platform open to actors, influencers, and other creative professionals. Talent accounts are self-service: users register directly through the platform and go through a structured onboarding flow before gaining access to their dashboard.

Talent exists in one of two subscription states:

- Free Tier — Limited monthly agency applications (limit TBD, between 1-3 per month), basic analytics, image upload restrictions, standard comp card download (PDF), hidden from agency discovery until profile is complete. Redirect

applications from agency links are always accepted regardless of monthly application limits.

- Studio+ Tier — Full feature access: unlimited agency applications, featured placement on the Pholio landing page, personalized match scores with agencies and boards, AI-driven recommendations (bio refinement, photo selection), smart portfolio builder, customizable comp card (layout, typography, colors), one-page dynamic talent website (hostable at talent-name.pholio.studio or a custom domain), advanced analytics including website traffic statistics, and no storage restrictions.

3.2 Agencies (Demand Side)

Agencies are the demand side of the marketplace. Agency accounts are not self-service — agencies must contact Pholio directly to be granted access. This gatekeeping is intentional: it ensures platform quality and positions Pholio as a curated environment rather than an open directory.

Once onboarded, agencies receive dashboard access at no cost. There is no subscription or fee structure for agencies. The agency's value is in the quality and volume of talent they can discover and pipeline through the platform.

4. Talent Onboarding Flow

4.1 Design Philosophy

The onboarding experience is the first impression Pholio makes on talent. It is designed as a cinematic, one-question-at-a-time conversational flow — not a traditional multi-field registration form. Each screen is fullscreen with a single focused prompt, black background, large serif typography, and a gold accent. The experience is intentionally more editorial than SaaS.

4.2 Phase 1: Account Creation

The onboarding begins with account creation, branching based on the method of authentication:

EMAIL / PASSWORD REGISTRATION

- Collects: First name, last name, email address, password, gender.

GOOGLE OAUTH REGISTRATION

- Name and email are pre-populated from the Google account. Password step is skipped.
- The only required input from the user is gender — the one field not provided by Google's OAuth payload.

INSTAGRAM OAUTH REGISTRATION

- Follows the same pattern as Google OAuth. Name, email pre-populated; gender is the only manual input.

4.3 Phase 2: AI-Powered Photo Analysis & Measurements

After account creation, the user uploads a single photo. This triggers the Groq Vision model (Llama-3.2-11b-Vision-Preview), which analyzes the image and extracts:

- Physical attributes: Hair color, eye color, body build/type.
- Predictive measurement estimates: Height, bust, waist, hips.

The platform then presents a measurements input screen pre-filled with the AI's estimates. Users are explicitly informed these are AI predictions, not confirmed values. They may confirm the estimates or manually adjust any field. This design achieves two goals simultaneously: it reduces friction (users don't start from a blank form) and collects accurate self-reported data by anchoring users to a reasonable starting point.

4.4 Phase 3: Clarifying Questions

After measurements, the onboarding collects two additional data points through the same conversational single-question format:

- Location: Where the talent is based (city/country).
- Experience Level: The talent's prior industry experience.

4.5 The Reveal Screen

Upon completing onboarding, users are directed to a /reveal screen — the emotional payoff of the onboarding experience. This screen presents:

- Radar Chart: A visual representation of the talent's fit across five modeling categories — Runway, Editorial, Commercial, Lifestyle — showing relative strength in each based on the collected data.
- Personalized Narrative: A brief AI-generated summary of the talent's profile and potential positioning.
- Action Plan: A structured list of first tasks to complete in the dashboard, setting clear expectations for what comes next.

4.6 Post-Onboarding: Profile Completion Gate

After the reveal screen, users enter their talent dashboard. The dashboard enforces a profile completion gate: talents are hidden from agency discovery until all core profile fields are filled. A persistent banner at the top of the dashboard communicates this status and the number of remaining incomplete sections.

4.7 Redirect Application Flow

Pholio supports a referral/redirect mechanism for agency partners. When an agency embeds a Pholio redirect link on their own website or in outreach, clicking that link routes a prospective talent through Pholio's onboarding. Upon completing their profile, the platform automatically detects the originating agency and prompts the talent to submit an application to that agency. This redirect-triggered application bypasses the standard monthly application limit — it is always accepted. This mechanism serves as a top-of-funnel tool for agency partners to drive qualified applicants through their own channels.

5. Talent Dashboard

5.1 Navigation Structure

The talent dashboard uses a top navigation bar with five primary sections: Overview, Profile, Portfolio, Analytics, and Applications. The Studio+ upgrade button is persistently visible in the navigation.

5.2 Overview Tab

The Overview tab is the talent's home screen. It is structured around three functional areas:

AT A GLANCE METRICS

- Profile Views — total views of the public talent profile.
- Downloads — number of times the talent's comp card has been downloaded.
- Applications — total number of agency applications submitted.
- Profile Strength — a percentage score reflecting profile completeness, used to guide action.

QUICK ACTIONS

- Download Comp Card — generates and downloads the talent's comp card as a professional PDF.
- Share Profile — copies the public profile URL to clipboard.
- View Public Profile — opens the public-facing profile as agencies would see it.

NEXT STEPS

A dynamic checklist of outstanding profile tasks (e.g., 'Complete your profile,' 'Upload portfolio images'), each with a direct action button. This persists until the talent is fully onboarded.

RECENT ACTIVITY

A feed of recent platform events relevant to the talent (application status changes, agency invitations, profile views).

5.3 Profile Tab

The Profile tab is the talent's core data management page. It is organized into labeled subsections accessible via a left-side navigation:

- Personal Details — First name, last name, and primary contact information visible to agencies.
- Heritage & Background — Ethnicity, nationality, and background information.
- Physical Attributes — Measurements (height, bust, waist, hips), eye color, hair color, build.

The profile features a large hero image at the top displaying the talent's name and location (e.g., 181 CM • DUBAI, UAE), mirroring the aesthetic of an agency's own website profile pages.

5.4 Portfolio Tab

The Portfolio tab manages the talent's uploaded images. Free tier users have a restricted image upload limit. Studio+ users have no storage restriction. The smart portfolio builder (Studio+) uses the Groq inference model to analyze uploaded images and auto-suggest which photos to feature on the comp card, based on image quality and category fit.

5.5 Analytics Tab

Free tier users receive basic analytics (profile views, downloads, application counts). Studio+ users receive advanced analytics including website traffic statistics for their talent.pholio.studio subdomain or custom domain, detailed breakdown of who has viewed the profile, and trend data over time.

5.6 Applications Tab

The Applications tab provides a complete view of all agency applications submitted by the talent — including status (Pending, Under Review, Accepted, Declined) and timestamps. Redirect-triggered applications are visually distinguished from standard applications.

6. Agency Dashboard

6.1 Access & Onboarding

Agencies cannot self-register. They must contact Pholio directly to request access. After a qualification conversation, Pholio manually provisions an agency account and provides login credentials. This friction is by design: it maintains the quality and exclusivity of the agency side of the platform.

There is no subscription cost for agencies. The agency dashboard is provided as a value-added discovery tool in exchange for the agency's participation in the marketplace.

6.2 Discover Tool

The Discover tool is the agency's primary mechanism for proactive talent scouting. Rather than passively waiting for applications, agents can search and filter the full Pholio talent pool using:

STRUCTURED FILTERS

- Location
- Height
- Age
- Eye Color
- Hair Color

NATURAL LANGUAGE SEARCH

Agents can enter free-text queries such as "5'10\" Redhead in NYC" and the platform returns relevant talent results. This is powered by the Groq inference model interpreting the query against the structured talent database.

INVITE ACTION

When an agent identifies a promising talent in the Discover view, they can send an Invite. This delivers a notification to the talent prompting them to apply to the agency. Invited profiles are automatically routed into the agency's application pipeline tagged with an "Invited" label, distinguishing them from organic applications.

6.3 Application Pipeline

The pipeline is the agency's applicant management system. It functions as a structured inbox organized into tabbed status buckets:

New	Incoming applications not yet reviewed. Zero-inbox workflow: agents aim to clear this tab.
Under Review	Applications flagged for consideration but not yet decided.
Accepted	Talent who have been signed or advanced. They appear on the agency's official roster within Pholio.

Declined	Rejected applications. Talent is notified. Record is preserved for history.
Archived	Applications removed from active view. Supports bulk-archive for pipeline hygiene.

6.4 Quick View Modal

Clicking any applicant opens a Quick View modal without navigating away from the pipeline list. The modal surfaces: measurements, bio, and portfolio photos — the core decision-making data for an agent. This design minimizes context switching and cognitive load during high-volume review sessions.

6.5 Applicant Actions

From the Quick View or full profile view, agents have four primary actions:

Accept	Moves application to Accepted tab. Talent receives an acceptance email notification. Profile appears on agency's Pholio roster.
Decline (Reject)	Moves application to Declined tab. Talent is notified. Record preserved.
Tag	Agents apply custom tags (e.g., 'Fall 2025', 'Runway Potential') for internal organization. Tags are not visible to talent.
Internal Notes	Private notes attached to a profile (e.g., 'Great look, needs updated walking video'). Not visible to talent.

6.6 Match Scores

The platform auto-calculates a percentage match score for each applicant relative to an agency's requirements or a specific board's criteria. For example, an agency that exclusively signs talent over 5'9" will see a low match score for a 5'5" applicant, allowing them to prioritize review efficiently. Because agencies frequently manage multiple boards with distinct requirements (e.g., Runway board vs. Commercial board), match scores may be calculated at the board level rather than the agency level. This behavior is currently under architectural review.

6.7 Bulk Operations

Agents can bulk-select multiple applicants to apply archive or decline actions simultaneously, supporting efficient pipeline hygiene and enabling a zero-inbox workflow for the New tab.

7. Comp Card & Studio+ Features

7.1 Comp Card

The comp card is the standard industry deliverable for talent representation — a one-page professional summary of a model's key stats, measurements, and photos used when submitting to agencies or productions. Pholio generates this programmatically via Puppeteer and exports it as a PDF.

Free tier users receive a standard comp card in a fixed template. Studio+ users receive full customization control over the comp card: layout structure, typography choices, and color scheme.

7.2 Studio+ Feature Suite

Studio+ is Pholio's premium subscription tier for talent. It unlocks the following capabilities:

- Unlimited Agency Applications — removes the monthly application cap entirely.
- Customizable Comp Card — full control over comp card layout, typography, and colors.
- Dynamic Talent Website — a one-page hosted website at talent-name.pholio.studio or a custom domain.
- Featured Placement — the talent is featured in the dedicated Featured section on the Pholio landing page.
- Personalized Match Scores — AI-calculated compatibility scores with specific agencies and boards.
- AI Recommendations — suggestions for bio refinement, photo selection optimization, and portfolio curation.
- Smart Portfolio Builder — AI analysis of uploaded images to auto-suggest which photos to feature on the comp card.

- Advanced Analytics — website traffic statistics, detailed profile view breakdowns, and trend data.
- No Storage Restrictions — unlimited image uploads.

8. Design System

8.1 Brand Philosophy

Pholio's visual design language is luxury-editorial. The aesthetic draws from high-end fashion publishing rather than conventional SaaS design patterns. The reference point is "Instagram meets Vogue" — sophisticated, minimal, and image-forward. Every design decision reinforces the premium positioning of the platform.

8.2 Color Palette

Primary Gold	#C9A55A — Primary actions, highlights, progress indicators, brand accents. Hover: #b08d45.
Dark Text	#0f172a — Headings, primary body text, important labels.
Slate Text	#64748b — Secondary text, descriptions, hints, placeholders.
Light Text	#94a3b8 — Tertiary text, disabled states.
Primary Background	#faf9f7 — Main page background.
Secondary Background	#f5f4f2 — Section backgrounds.
Card Background	#ffffff — Card surfaces.
Background Gradient	linear-gradient(135deg, #faf9f7 0%, #f5f4f2 100%)
Light Border	#e2e8f0
Hover Border	#cbd5e1
Onboarding Background	#000000 — Full black for the cinematic onboarding experience.

8.3 Typography

Primary font family: Inter, with system font fallbacks. Font smoothing applied via -webkit-font-smoothing: antialiased. Type scale:

H1	2rem (32px) | Weight 700 | Letter-spacing -0.03em
H2	1.875rem (30px) | Weight 600 | Letter-spacing -0.02em
H3	1.5rem (24px) | Weight 600
H4	1.25rem (20px) | Weight 600
Body Large	1.05rem (16.8px) | Weight 400 | Line-height 1.5
Body Base	1rem (16px) | Weight 400 | Line-height 1.6
Body Small	0.9rem (14.4px) | Weight 400
Label (Tiny)	0.75rem (12px) | Weight 500 | Uppercase | Letter-spacing 0.08em

Onboarding screens use a mixed serif/sans-serif typographic treatment: key prompt words are rendered in an italic serif font (gold color) while the surrounding question text uses the standard Inter weight. This contrast creates the editorial, cinematic feel of the onboarding.

8.4 Spacing System

Base unit: 1rem (16px). Component-specific spacings: form fields spaced at 1.75rem (28px); section padding at 3rem horizontal / 2rem vertical; card padding between 2.5rem and 3.5rem depending on context; button padding at 0.875rem 1.75rem.

8.5 Border Radius

Small	4px — badges, small elements
Medium	8px — inputs, buttons, cards
Large	12px — upload areas, large cards
Extra Large	16px — main containers, hero cards

8.6 Elevation & Shadows

Card Shadow	box-shadow: 0 20px 40px -10px rgba(0, 0, 0, 0.08)
Button Hover Shadow	box-shadow: 0 4px 12px rgba(201, 165, 90, 0.2)
Focus Ring	box-shadow: 0 0 0 4px rgba(201, 165, 90, 0.1)

8.7 Button Styles

Primary Button	Background: #C9A55A | Color: white | border-radius: 8px | font-weight: 500 | min-width: 140px. Hover: background #b08d45, translateY(-1px), gold shadow.
Secondary Button	Background: transparent | Color: #64748b | border-radius: 8px. Hover: color #0f172a, background #f1f5f9.
Disabled State	opacity: 0.6 | cursor: not-allowed | no transform | no shadow.

9. Core Data Model

9.1 Key Entities

The following entities represent the core data structure of the Pholio platform stored in Neon PostgreSQL:

- users — Core authentication record. Links to Firebase Auth UID. Stores role (talent | agency), subscription status, and created_at.
- talent_profiles — Full talent data: name, gender, location, experience level, physical attributes (height, measurements, eye color, hair color), bio, profile completion status, and profile_strength score.
- portfolio_images — Individual image records linked to a talent. Stores storage URL, upload timestamp, AI-generated attributes from vision analysis, and whether the image is featured on the comp card.
- agencies — Agency account records: name, contact information, onboarding date, and internal Pholio status.
- applications — Tracks every talent-to-agency application. Fields: talent_id, agency_id, status (new | under_review | accepted | declined | archived), source (organic | redirect | invited), applied_at, reviewed_at.
- agency_tags — Custom tags applied by agents to applications (e.g., 'Fall 2025'). Not visible to talent.
- agency_notes — Private agent notes attached to a specific application. Not visible to talent.
- invitations — Records of agency Invite actions sent to talent from the Discover tool.
- redirect_links — Tracks agency-specific redirect URLs for the top-of-funnel redirect application flow.

10. Key User Flows Summary

10.1 Talent: First-Time Registration to Discoverable Profile

Step 1	Select auth method (Email, Google, Instagram).
Step 2	Complete Phase 1 inputs: name (if not OAuth), email (if not OAuth), password (if not OAuth), gender.
Step 3	Upload one photo. AI vision model analyzes image, extracts attributes and measurement estimates.
Step 4	Review and confirm/adjust pre-filled measurements.
Step 5	Answer clarifying questions: location, experience level.
Step 6	View /reveal screen: radar chart, narrative, action plan.
Step 7	Enter talent dashboard. Profile completion gate active. Banner shows remaining sections.
Step 8	Complete remaining profile sections. Profile becomes visible to agencies in Discover.
Step 9 (if redirect)	Platform detects originating agency. Prompts talent to submit application. Application is processed regardless of monthly limit.

10.2 Agency: Proactive Discovery to Signed Talent

Step 1	Log in to agency dashboard (credentials provisioned by Pholio).
Step 2	Navigate to Discover tool. Apply filters or enter natural language search query.
Step 3	Review talent cards. Click Invite on promising profiles.
Step 4	Invited talent appears in Applications pipeline tagged as Invited.
Step 5	Open Quick View modal to review measurements, bio, photos.
Step 6	Apply action: Accept / Decline / Tag / Add Note.
Step 7 (Accept)	Talent notified via email. Profile appears on agency's Pholio roster. Relationship moves off-platform.

11. Development Status & Roadmap Notes

11.1 Current Status

Pholio is currently in active development and has not yet launched publicly. The core talent onboarding flow, talent dashboard, agency dashboard, and application pipeline are implemented. The platform is in a pre-launch testing state with the US market as the initial target.

11.2 Known Open Items

- Agency Boards: The multi-board architecture within a single agency account is not fully specified. The current implementation uses agency-level settings for match score calculation, but the intention is to support board-level criteria (e.g., Runway board vs. Commercial board with different requirements). This is under architectural review.
- Comp Card Structure: The exact layout and content structure of the comp card (beyond PDF export) is still being finalized. Free vs. Studio+ template differentiation is defined; the specific design templates are in progress.
- Application Limits: The exact free-tier monthly application limit has not been finalized. The target range is 1-3 applications per month.
- Boards Feature (Agency): The Boards system for agency-side organization is listed as "in development." Currently agents use Tags for categorization.

11.3 Revenue Model

The sole current revenue stream is the Studio+ subscription, processed through Stripe. Pricing is not detailed in this document. The agency side is provided free of charge as a supply-side acquisition strategy — the value to Pholio is participation quality and the ability to sell the platform's discovery capability to talent as a reason to upgrade to Studio+.

Appendix: Terminology Reference

Comp Card	A one-page professional PDF summarizing a talent's key stats, measurements, and photos. The standard industry submission format.
Redirect Application	An application triggered by a talent arriving at Pholio via an agency's embed link. Bypasses standard monthly application limits.

Profile Strength	A percentage score reflecting profile completeness. Talent with incomplete profiles are hidden from agency discovery.
Invite	An agency-initiated action in the Discover tool that sends a notification to a talent encouraging them to apply.
Quick View	A modal UI pattern in the agency pipeline allowing agents to review a talent's key data without leaving the applicant list.
Match Score	An AI-calculated percentage indicating how closely a talent's attributes match an agency's or board's stated requirements.
Studio+	Pholio's premium talent subscription tier. Unlocks unlimited applications, customization, AI tools, analytics, and featured placement.
Reveal Screen	The post-onboarding screen presenting the talent's radar chart, narrative summary, and action plan.
Natural Language Search	Free-text query input in the Discover tool interpreted by the Groq inference model to return relevant talent results.